The Cooper Intelligence Technical Report: Template and Style Guide

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Abstract

This document is at once the reference example of the Cooper Intelligence technical-report class and its usage guide. It documents what the template fixes, namely the identity block, the typographic grid and the metadata hooks, and what it deliberately leaves to authors, namely the argument, the evidence and the length. Every element shown here can be copied verbatim into a new report: front matter, numbered sections, figures, tables, displayed mathematics, citations, an appendix and the copyright line.

Keywords: technical report; house style; typography; template; release process

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1. Purpose and Scope

Cooper Intelligence publishes technical reports for two audiences at once: clients who need a defensible account of a design or a method, and internal teams who need to reuse the reasoning later. When every team formats its own PDF, the organisation ends up with documents that look like they come from different companies. This template exists to remove that variance without touching the substance of what teams write.

The template is published under the guidance of Boogon, which owns the release review. A report that leaves the organisation therefore passes two gates: the author's technical review, and a Boogon editorial check for conformance with the conventions described here.

1.1. What the template fixes

Three things are fixed, and authors should not override them in individual reports:

- **Identity.** The wordmark, organisation name, corresponding contact and the copyright line. These are set once through the branding hooks in Section 2.
- **Grid.** A4 paper, a single column, 2.5 cm side margins and 3 cm top and bottom margins. The measure lands near 90 characters, which is wide enough for tables and narrow enough for sustained reading.
- **Running furniture.** The short title in the running header, the page number in the footer, and the grey rules above and below the text block.

1.2. What the template leaves open

Section structure, length, tone, the choice between prose and bullet lists, and the amount of mathematics are editorial decisions. The template provides machinery, not a house argument. A three-page note and a thirty-page study use the same class and should both look deliberate.

1.3. Scope of the current release

This release covers English-language reports. Chinese-language typesetting is out of scope for the present version: the class is built on `extarticle` with a Latin font stack, and a credible Chinese edition requires a different font and line-breaking configuration rather than a translation of this document.

Reports produced with the template are distributed as PDF, always accompanied by their LaTeX sources. The sources matter because a report is expected to remain reproducible years after its publication, when neither the author nor the original tooling is available.

2. Document Anatomy

A Cooper report has exactly two page types: a first page that carries the identity block, and running pages that carry only furniture. Understanding the split is enough to typeset a report correctly.

The class itself is a fork of the typesetting layer used for the DeepSeek-V3 technical report (DeepSeek-AI, 2024), with the identity block replaced and the running header activated. The debt is deliberate: that layout was already proven on the page sizes and print runs Cooper cares about.

2.1. The first page

The header holds the wordmark on the left and the ISO-8601 date on the right (2026-10-28). Below the header rule the title is set in bold at 16 pt, followed by the author block and the abstract. The footer carries the copyright line at the right and, when the corresponding author is declared, its address at the left.

The abstract is declared before `\begin{document}`. This is unusual but deliberate: the class rewrites the `abstract` environment so that its content can be placed on the title block without floating, and the declaration has to happen before the document body starts.

2.2. Running pages

From page two onwards the header is a single centred line, the short title, in 11 pt. The footer carries the page number, also centred. Both are separated from the text block by a 1 pt rule drawn in grey, which keeps the rule visible on screen without competing with body text on paper.

2.3. Metadata hooks

All organisation-specific strings are collected in five macros, so that a rebrand never touches the body of a report. Table 1 lists them together with the class options that switch the optional parts on.

Table 1 | The branding hooks and the class options that activate optional elements.

Hook or option	Rendered where	Default
`\reportorg`	copyright line	Cooper Intelligence
`\reportshorttitle`	running header	Technical Report
`\reportlogo`	first-page header	none
`\reportaddress`	first-page footer, left	none
`\reportinternalnote`	footer when `internal`	Confidential. Internal Only.
`logo`	draws the wordmark	off
`copyright`	draws the copyright line	off
`address`	draws the address line	off
`internal`	marks the report as internal	off
`nonumbering`	suppresses section numbers	off

Options are passed on the `\documentclass` line. A released report enables `logo` and `copyright`; an internal draft adds `internal`, which prints the confidentiality note in the footer.

3. Typographic System

The template uses one text family, one matching mathematics family and one monospaced family. That is the whole system; there is no second accent colour and no decorative element other than the wordmark.

3.1. Type families

Body text is set in Bitstream Charter. Charter was drawn for low-resolution printing and holds up unusually well at 11 pt on coated paper, which is the medium most Cooper reports end up in. Mathematics is taken from the `newtx` companion set configured for Charter, so that symbols in a display equation do not change colour or weight relative to the surrounding prose. Monospaced text, used for file names, macros and code, comes from Latin Modern Typewriter scaled up by ten per cent to match Charter's x-height. This pairing follows the standard LaTeX practice of matching a text face with the mathematics set drawn for it (Mittelbach and Goossens, 2023).

3.2. Sizes and spacing

Table 2 records the sizes the class applies. Authors normally never set a size by hand: the class assigns one to each structural role, and a report that respects the roles will look like every other report.

Table 2 | Type sizes by role. All values are as set by the class.

Role	Size	Notes
Title	16/20 pt, bold	ragged right, no automatic hyphenation
Author block	11/15 pt	affiliation line set at 13/15 pt, bold
Abstract	11/12 pt	paragraph measure, ragged bottom
Keywords	11/12 pt, italic	appended to the abstract block
Section heading	13/14 pt, bold	number, then 0.5 em gap
Subsection	body size, bold	number, then 0.5 em gap
Subsubsection	body size, bold italic	run-in numbering
Running header	11/12 pt	short title only
Copyright	7/9 pt	first-page footer, right

Vertical space follows the block structure rather than the page: 0.5 baselineskip between paragraphs, and 5 pt below a section heading. The grid is fixed at 2.5 cm side margins with 3 cm at the top and bottom, and the header cell is 40 pt tall, which is what allows the 138 pt wordmark to sit on the first page without pushing the title down.

3.3. Rules and captions

Header and footer rules are drawn at 1 pt in grey, on the first page and on running pages alike. Captions use the pipe separator, so a caption reads *Figure 1 | The release path for a Cooper technical report*; the separator is a property of the class, not something an author types.

4. Figures, Tables and Displayed Mathematics

The three element types below cover most reports. They are shown once, correctly, so that a new author can copy the pattern rather than reconstruct it.

4.1. Figures

Figures are centred, float by default, and carry a caption below. Vector output is expected: a PDF produced by a plotting tool, or a diagram drawn in Ti*k*Z as in Figure 1. Raster images are

acceptable only for screenshots, and then at a resolution that survives print.

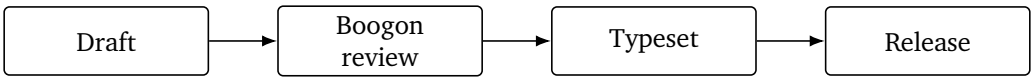


Figure 1 | The release path for a Cooper technical report. Editorial review sits between drafting and typesetting, so that a report is never typeset twice for the same objection.

4.2. Tables

Tables use `booktabs` rules and no vertical lines; the caption goes above. Numbers are aligned on the decimal, and units are stated in the column head, never repeated in every cell. Table 3 is the smallest complete example.

Table 3 | Conformance of a report to the template, by section, for a recent review cycle.

Section	Checks run	Conformant (%)
Front matter	14	100.0
Body sections	63	96.8
Figures and tables	22	90.9
References	9	100.0
Total	108	96.3

4.3. Displayed mathematics

Equations are numbered, and every symbol is defined in the surrounding prose. A conformance rate, for instance, can be written as

$$C = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\left[\, d_i \le \tau \,\right], \tag{1}$$

where d_i is the deviation of check i from its expected value, τ is the tolerance agreed during review, and $\mathbf{1}[\cdot]$ is the indicator function. Equation (1) is quoted in the review summary; it is also the quantity that Table 3 reports in percent.

Inline mathematics follows the same font. A tolerance of $\tau = 0.05$ is typical for typographic checks. The treatment of floats and cross-references used here follows the standard account in Lamport (1994).

5. Composition Recipes

A new report starts by copying `sample/` and editing three blocks. Everything else in this file can be deleted without breaking the class.

5.1. Front matter

The document class line and the branding hooks are the only places where organisation-level information appears. A released report sets them as follows.

```
\documentclass[11pt, a4paper, logo, copyright]{boogonreport}

\renewcommand{\reportorg}{Cooper Intelligence}
\renewcommand{\reportshorttitle}{Cooper Intelligence Technical Report}
\renewcommand{\reportlogo}{../logo/cooper-logo.pdf}

\title{\centering Your Report Title}

\author[*]{Cooper Intelligence\\
  \small under the guidance of Boogon\\
  \small \texttt{official@boogon.com}}

\keywords{three to six terms, semicolon separated}

\begin{abstract}
  One paragraph. It states the question, the method and the result,
  in that order, and stops.
\end{abstract}
```

5.2. Sections as separate files

Long reports keep each section in its own file so that two authors never edit the same file. The main file then only sequences them, which also makes the order of sections a one-line decision during review.

```
\input{sections/01-purpose}
\input{sections/02-anatomy}
\input{sections/03-typography}
```

5.3. The contents page

Reports of more than a few pages carry a contents page, as the upstream report this class derives from does. The class provides one command for it: `\reporttoc`, used directly after `\maketitle`. It starts a new page, sets the contents at 0.9 line spacing so that a long report still fits them on one page, and starts another new page afterwards.

```
\maketitle
\reporttoc
```

The contents list sections and subsections; deeper levels are excluded by default, which keeps the page scannable. Page numbers and hyperlinks are generated by LaTeX, so the document is built twice, as usual, before the numbers settle.

5.4. Citations and bibliography

Citations use `natbib` with author–year output, so that the argument can name a source in running text. The bibliography style is `plainnat`; references are kept in `refs.bib` next to the main file.

```
\citep{lamport1994}    % (Lamport, 1994)
\citet{mittelbach2023} % Mittelbach and Goossens (2023)

\bibliographystyle{plainnat}
\bibliography{refs}
```

5.5. Building

The document is built with `latexmk`, which runs the required number of LaTeX passes and invokes BibTeX when the bibliography changes. A clean build from the repository root is one command.

```
latexmk -pdf sample/main.tex
```

Build artefacts (`.aux`, `.log`, `.pdf`) are ignored by the repository; only sources are committed. A report is shipped as the pair of PDF and sources, never as a PDF alone.

6. Release Practice

A report is finished when it can be re-typeset from its sources by someone who was not involved in writing it. That test drives the release rules below.

6.1. Review

Release review is held by Boogon. The reviewer checks the document against the template conventions, not against the argument: correct class, branding hooks set, figures vectorial, tables free of vertical rules, every reference resolved, copyright line present. A report that fails these checks is returned once, with a list, rather than line-edited by the reviewer.

6.2. Versioning and archival

Each report is archived with four artefacts: the PDF, the LaTeX sources, the bibliography and the logo assets it referenced. The archive is immutable; a correction produces a new version, and the version is recorded in the report's file name and cover note, since the class deliberately has no version field in the title block.

Template changes are released separately from reports. Because a report pins the class it was built with, a change to the template does not silently alter an already published document.

6.3. When the template changes

A change to the class file requires two things: this sample must still build without modification, and any new hook must be documented in Section 2. If a change alters how an existing report renders, it is a breaking change and the previous class file is kept alongside the new one until the archived reports no longer need it.

References

DeepSeek-AI. DeepSeek-V3 technical report. arXiv preprint arXiv:2412.19437, DeepSeek-AI, 2024. The typographic design of this class derives from the report's LaTeX source.

Leslie Lamport. *LaTeX: A Document Preparation System*. Addison-Wesley, Reading, MA, 2nd edition, 1994.

Frank Mittelbach and Michel Goossens. *The LaTeX Companion*. Addison-Wesley, Boston, 3rd edition, 2023.

A. Pre-flight Checklist

Run through this list before a report goes to Boogon review. Every item has a concrete observation, so two reviewers reach the same verdict.

1. The `\documentclass` line carries `a4paper`, `logo` and `copyright`; `internal` is present only on drafts.
2. The wordmark on the first page is the current release, not a screenshot.
3. The author block names Cooper Intelligence, the Boogon guidance line and the contact address.
4. The abstract states question, method and result, and fits without spilling onto the second page.
5. Section headings are numbered and appear in the order of the argument.
6. Every figure is vectorial, centred and captioned below; every table captioned above and free of vertical rules.
7. Every equation is numbered and every symbol it introduces is defined in prose.
8. Every citation resolves; the bibliography builds without warnings.
9. The copyright line reads © 2026 Cooper Intelligence. All rights reserved.
10. The PDF and the sources are archived together, and the PDF was produced from exactly those sources.